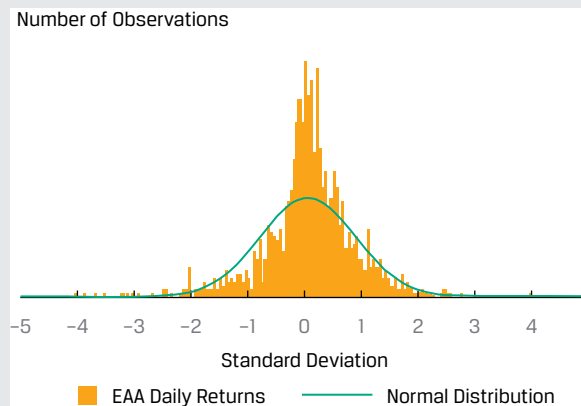


The returns reflect negative skewness, which is illustrated in Exhibit 21 by comparing the distribution of the daily returns with a normal distribution with the same mean and standard deviation.

#### Exhibit 21: Negative Skewness



Using both the statistics and the graph, we see the following:

- The distribution is negatively skewed, as indicated by the negative calculated skewness of  $-0.4260$  and the influence of observations below the mean of  $0.0347$  percent.
- The highest frequency of returns occurs within the  $0.0$  to  $0.5$  standard deviations from the mean (i.e., the mode is greater than the mean as the positive returns are offset by extreme negative deviations).
- The distribution is fat-tailed, as indicated by the positive excess kurtosis of  $3.7962$ . In Exhibit 21, we can see fat tails, a concentration of returns around the mean, and fewer observations in the regions between the central region and the two-tail regions.

To understand the trading liquidity of a stock, investors often look at the distribution of the daily trading volume for a stock. Analyzing the daily volume can provide insights about the interest in the stock, what factors may drive interest in the stock as well as whether the market can absorb a large trade in the stock. The latter may be of interest to investors interested in either establishing or exiting a large position in the particular stock.

#### INTERPRETING SKEWNESS AND KURTOSIS

Consider the daily trading volume for a stock for one year, as shown in Exhibit 22. In addition to the count of observations within each bin or interval, the number of observations anticipated based on a normal distribution (given the sample arithmetic average and standard deviation) is provided as well. The average trading volume per day for the stock during the year was  $8.6$  million shares, and the standard deviation was  $4.9$  million shares.